SOPHIA SCHIFFER

sophia.engineer2024@gmail.com • 425.209.6703 • sophia-schiffer.github.io

3900 N Pine Grove Ave, Apt. 502, Chicago, IL 60613

EDUCATION

Northwestern University **Evanston, Illinois**
Bachelor of Science in **Mechanical Engineering** June 2024

GPA: 3.740/4.000

EMPLOYMENT & PROJECTS

Focal Point, LLC **Chicago, Illinois**
Mechanical Engineer I – Custom Design November 2024 – Present
•Design custom aluminum housings and lenses to match customer-specified patterns for all Focal Point's linear product lines (e.g. Seem 1, Seem 2, Focus, etc. with various suspended or recessed mounting types)
•Compile bills of materials for custom lighting orders to incorporate designed components, LEDs and drivers for specified lumen outputs, and special requests for mounting, sensor, optic, and finish offerings

Mechanical Engineering Internship Projects June 2023 – August 2023
•Developed step light for hospital room, meeting even distribution and low 40-lumen output requirements
•Prototyped and developed ceiling cover shrouds for recessed narrow-beam downlights with high tilt angles up to 40°

Interactive & Emergent Autonomy Lab **Evanston, Illinois**
Active Control for WALTER Soft Robot July 2022 – June 2024
•Tuned primitive walking gaits for automated tree-search learning of legged locomotion with 4-minute training time
•Implemented RRT-based path planner for obstacle navigation in simulation
Publication: Ketchum, Jake, Sophia Schiffer, Muchen Sun, Pranav Kaarthik, Ryan Landon Truby and Todd D. Murphey. "Automated Gait Generation For Walking, Soft Robotic Quadrupeds." *ArXiv* abs/2310.00498 (2023): n. pag.

Autonomous Pin Shooter Capstone Project **Evanston, Illinois**
Systems Integrator November 2023 – December 2023
•Designed robust AprilTag tracking pipeline for detecting precise placement of Nerf Modulus Blaster
•Implemented MoveIt2 motion planning to grasp and place Nerf guns with Franka Emika Panda robot

Northwestern-Argonne Institute for Science and Engineering (NAISE) **Lemont, Illinois**
3D Printed Silicone Soft Robotic Gripper June 2021 – August 2021
•Developed code for Arduino-based control system for pneumatically actuated soft finger
•Conceptualized 2^{nd} iteration 3-finger gripper using inspiration from octopus anatomy

Griffin Museum of Science and Industry **Chicago, Illinois**
Guest Engagement Facilitator I June 2024 – November 2024
•Provided informal education and demos to museum guests on storms, electricity, and technology
•Led guided tours of the U-505 (WWII German U-boat) and coal mine, explaining both history and machinery

SKILLS

- **Programming:** Python, MATLAB, C, C++, ROS/ROS2, HTML, JavaScript
- **Engineering Software:** SOLIDWORKS, AutoCAD, Fusion360, Ansys (FEA), Arduino, Siemens NX, Oracle
- **Shop:** Band Saw, CNC, Drill Press, Lathe, Mill, 3D Printer, Sand Casting, Sewing Machine
- **Design:** Design for Manufacturing, Injection Molding, Sheet Metal, Design for Aesthetics, LED Lighting
- **Creative Software:** Inkscape, Adobe Photoshop, Digital Audio Workstations (FL Studio), Canva